

## Dividing fractions



**1** Simplify:

a  $\frac{1}{8} \div \frac{1}{5}$

e  $\frac{1}{10} \div \frac{1}{3}$

b  $\frac{1}{2} \div \frac{9}{10}$

f  $\frac{1}{2} \div \frac{1}{9}$

c  $\frac{1}{5} \div \frac{3}{7}$

g  $\frac{2}{3} \div \frac{1}{8}$

d  $\frac{3}{4} \div \frac{1}{9}$

h  $\frac{1}{11} \div \frac{3}{7}$

**2** Simplify:

a  $\frac{2}{7} \div \frac{5}{3}$

e  $\frac{28}{9} \div \frac{4}{9}$

i  $\frac{2}{3} \div \frac{11}{15}$

m  $\frac{2}{36} \div \frac{4}{7}$

b  $\frac{9}{10} \div \frac{10}{9}$

f  $\frac{3}{15} \div \frac{15}{4}$

j  $\frac{10}{7} \div \frac{10}{7}$

n  $\frac{3}{55} \div \frac{7}{20}$

c  $\frac{2}{7} \div \frac{7}{18}$

g  $\frac{21}{27} \div \frac{3}{2}$

k  $\frac{4}{5} \div \frac{36}{35}$

o  $\frac{10}{99} \div \frac{45}{22}$

d  $\frac{3}{15} \div \frac{4}{3}$

h  $\frac{5}{7} \div \frac{45}{2}$

l  $\frac{3}{7} \div \frac{14}{27}$

p  $\frac{22}{27} \div \frac{3}{2}$

**3** Simplify:

a  $3 \div \frac{1}{11}$

e  $\frac{3}{5} \div 4$

b  $\frac{1}{9} \div 8$

f  $\frac{65}{9} \div 5$

c  $\frac{4}{3} \div 5$

g  $3 \div \frac{39}{10}$

d  $\frac{6}{11} \div 8$

h  $7 \div \frac{98}{12}$

**4** Simplify:

a  $1\frac{2}{5} \div \frac{3}{4}$

e  $2\frac{1}{3} \div 6\frac{2}{12}$

b  $2\frac{1}{4} \div 1\frac{2}{5}$

f  $3\frac{3}{8} \div 2\frac{1}{4}$

c  $1\frac{2}{7} \div 3\frac{1}{2}$

g  $6\frac{3}{4} \div 4\frac{2}{7}$

d  $3\frac{1}{3} \div 2\frac{1}{2}$

h  $8\frac{1}{2} \div 1\frac{5}{6}$

**5** Simplify:

a  $\frac{1}{5} \times \frac{1}{3} \div \frac{1}{2}$

e  $\frac{1}{7} \times \frac{2}{9} \div \frac{4}{27}$

b  $\frac{2}{3} \div \frac{1}{9} \times \frac{2}{5}$

f  $\frac{1}{7} \times \frac{5}{8} \div \frac{15}{49}$

c  $\frac{3}{6} \div \frac{12}{5} \div \frac{1}{2}$

g  $\frac{18}{25} \div \frac{9}{10} \times \frac{1}{3}$

d  $\frac{3}{4} \div \frac{2}{5} \times \frac{1}{6}$

h  $\frac{8}{11} \div \frac{2}{3} \times \frac{15}{2}$

## Applications

**6** A family drinks  $\frac{3}{4}$  of a carton of milk per day. How many days will 6 cartons of milk last the family?

- 7 A glass holds  $\frac{3}{8}$  of a litre of lemonade. How many glasses of lemonade can be filled with 6 litres of lemonade?
- 8 Yang eats  $\frac{2}{5}$  of a packet of biscuits each day. If he buys 12 packets, how many days will the packets last?
- 9 If  $\frac{2}{3}$  of a tart is shared equally between 5 people, what fraction of the tart would each person receive?
- 10 A truck tank contains 42 L when  $\frac{3}{4}$  full. What is the capacity of the tank when full?
- 11 If a garden snail can move  $\frac{1}{100}$  km/h, how many hours would it take for the garden snail to reach  $\frac{3}{4}$  km?
- 12 Christine has  $1\frac{3}{4}$  of a metre of ribbon. She uses  $\frac{7}{20}$  to wrap a present. How many presents can she wrap with her ribbon?
- 13 Sheldon is planning to sell banana bread on the upcoming school fair. If a loaf requires  $\frac{3}{4}$  cups of sugar, how many banana bread loaves can Sheldon make if he has  $8\frac{2}{8}$  cups of sugar on hand?
- 14 Jean is making bows for her gift wrapping business. A bow requires  $6\frac{1}{6}$  inches of ribbon. If she has  $55\frac{1}{2}$  inches of ribbon, how many bows can Jean make?
- 15 Benjamin wants to engrave his name on a metal plate. There is  $1\frac{2}{4}$  cm space on the plate. There are three sizes of letters to choose from:  $\frac{1}{8}$  cm wide,  $\frac{1}{6}$  cm wide and  $\frac{1}{4}$  cm wide.
- a Find the maximum number of letters that Benjamin can engrave on the plate using the following sizes:
- i  $\frac{1}{8}$  cm wide      ii  $\frac{1}{6}$  cm wide      iii  $\frac{1}{4}$  cm wide
- b Which letter sizes could he use for his name?